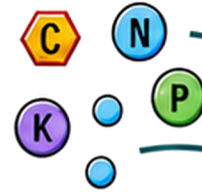
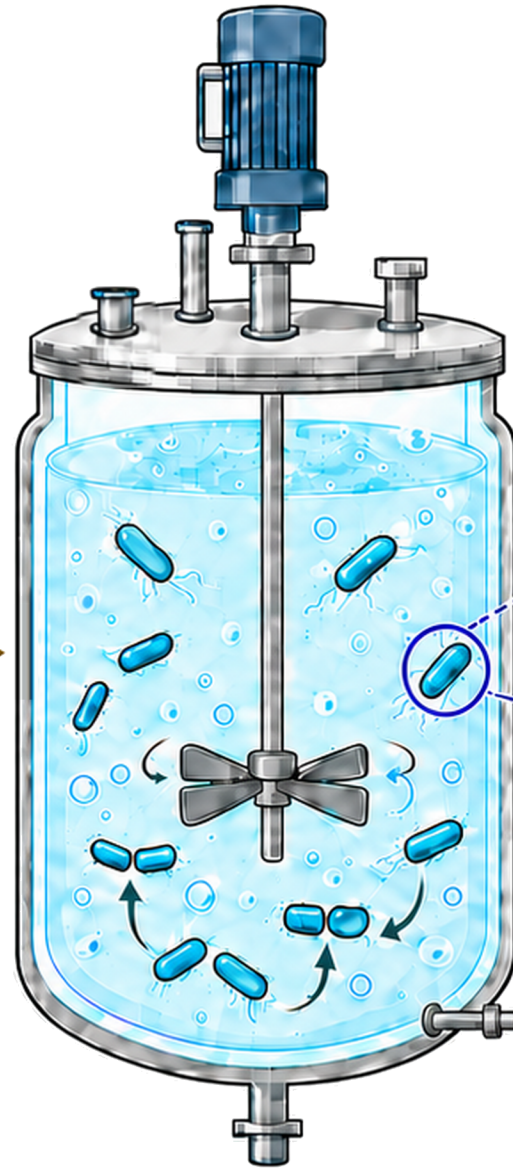




Side stream



Carbon, nitrogen
and minerals



Nutrient
uptake

Metabolism

ATP

Biomass
precursors

Secreted metabolites



Microbial biomass